

# Design of IOT Architecture for Battery Charging Management System

Dr. Sujith Kalluri<sup>1</sup>, Jagadeeswara Chandra bose.k<sup>1</sup>,  
Jaswanth Kumar<sup>2</sup>, Uttam Kumar<sup>2</sup>,  
<sup>1</sup>Department of ECE, SRM University AP



1<sup>st</sup> Edition of ECE Open Day  
School of Engineering and Sciences  
Department of Electronics and  
Communications Engineering,  
M. Tech Project 2023-2025

## Abstract

Rapid growing of electric vehicle (EV) adoption faces challenges such as range anxiety and inconsistent charging infrastructure, which can be effectively addressed through an IoT-based architecture that offers real-time monitoring, predictive analytics, and mobile charging units, ultimately enhancing user experience, optimizing energy management, and promoting sustainable energy solutions for a cleaner future.

## Introduction

### •Growing Demand for Charging Infrastructure:

The rapid increase in electric vehicle (EV) sales is creating a pressing need for more efficient and accessible charging stations to alleviate range anxiety among users.

### •Integration of Renewable Energy:

There is a critical requirement for sustainable charging solutions that incorporate renewable energy sources, such as solar power, to align with environmental sustainability goals.

### •Cost Reduction and Innovation:

The necessity to reduce operational costs in charging management is driving the adoption of innovative IoT technologies aimed at optimizing battery management and enhancing overall efficiency.

## Problem Statement

- Demand of Charging Infrastructure Expansion
- Grid Load Disturbances
- High Installation and Operational Costs
- Inefficient Load Management
- Limited Remote Monitoring

## Objectives

- Develop an IoT-based architecture for optimized battery charging management.
- Incorporate real-time data monitoring and predictive analytics.
- Enhance operational cost efficiency by up to 88.5%.
- Promote sustainable energy integration through solar and ESS.

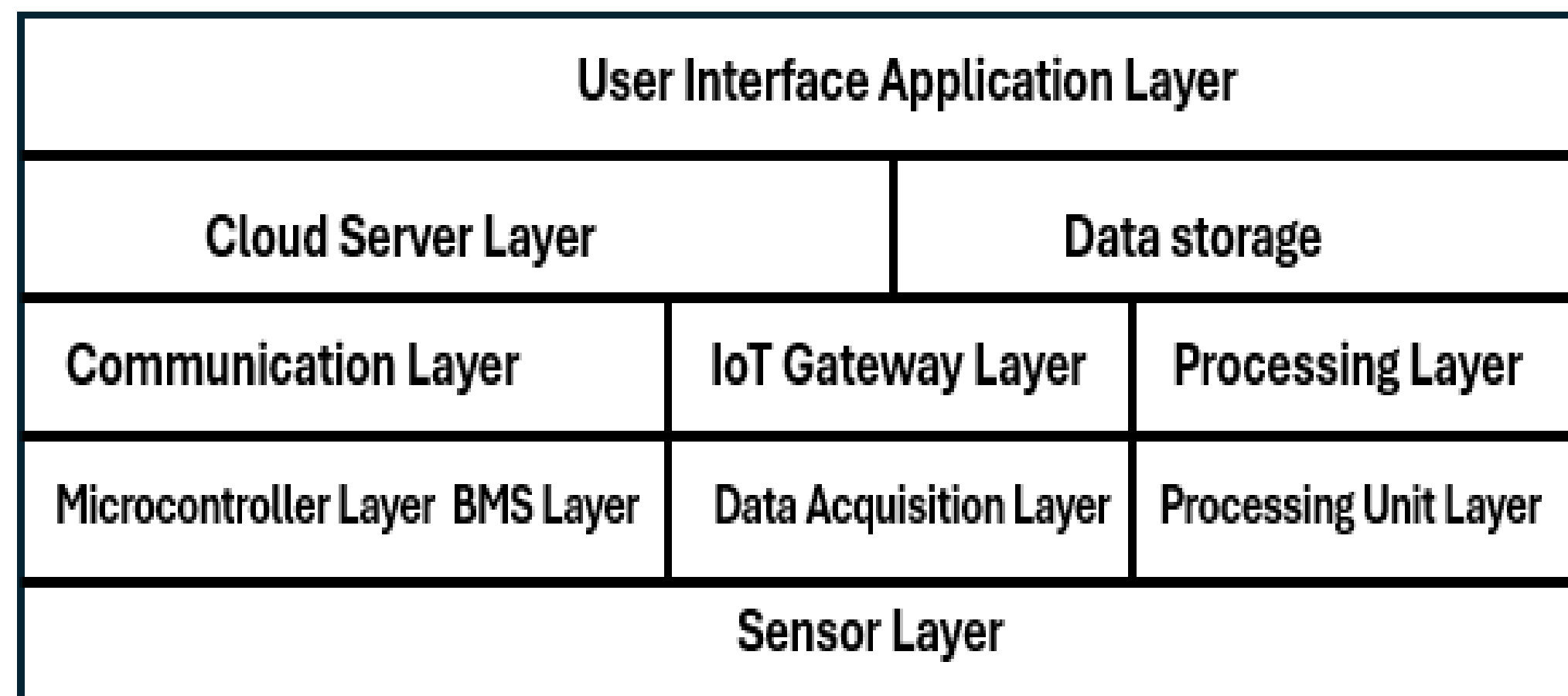
## Project Novelty

- Innovative IOT 5 layered Architecture.
- Integration of multiple energy sources.
- Central monitoring and controlling.
- Real-Time data processing and user interface.
- Cloud connectivity.
- Cost efficiency model.
- Scalability and Adaptability.
- Portable charging station.

## Methodology

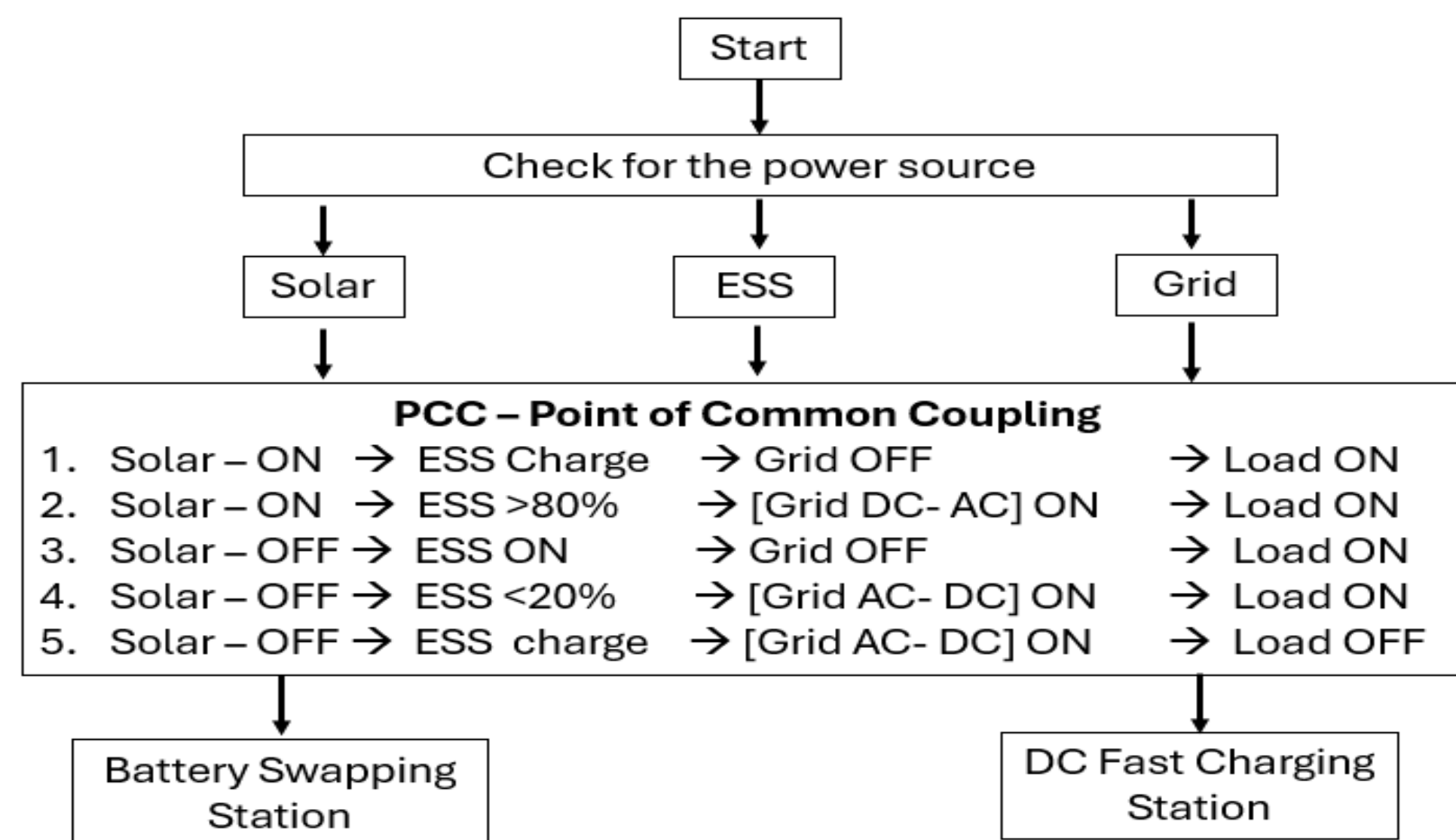
### IoT Five-Layered Architecture

- User Interface Application Layer
- Cloud Server and Data Storage
- Communication Layer
- Microcontroller Layer
- Sensor Layer.



### Point of Common Coupling (PCC)

Combines solar power, ESS, and grid power for



## Algorithms

### MPPT (Maximum Power Point Tracking) Algorithm:

Optimizes solar panel output by continuously adjusting the operating point to maximize energy capture under varying conditions.

### Battery Swapping Station Algorithm:

Manages logistics and scheduling for battery swaps, optimizing inventory and availability of charged batteries to improve service efficiency.

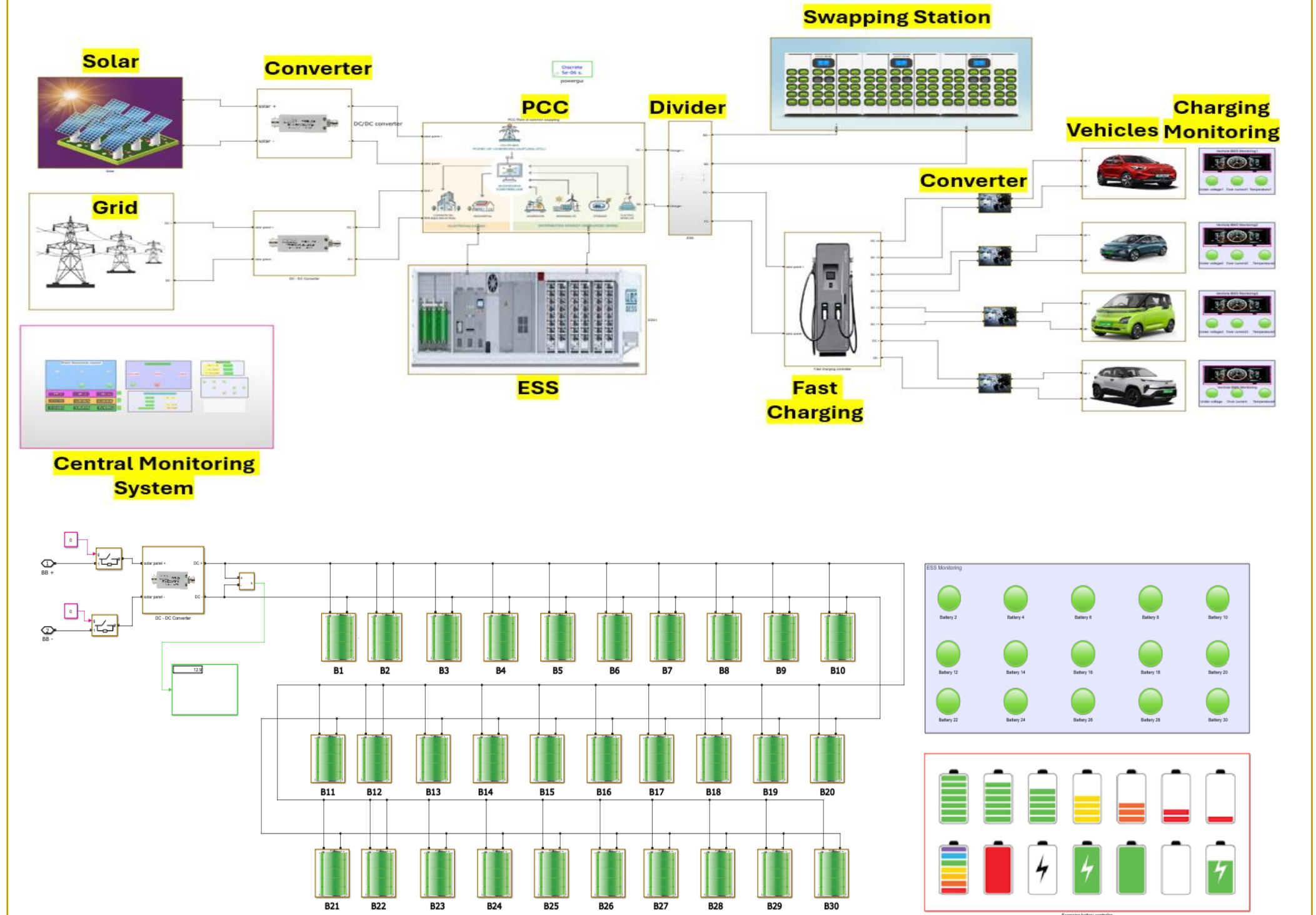
### Vehicle Charging Power Control Calculation:

Calculates optimal charging rates based on battery state-of-charge (SOC) and user demand, ensuring safe and efficient charging.

### Safety and Blast Condition Prediction using ML:

Utilizes machine learning models to analyze historical data and predict potential safety hazards in battery charging environments, enabling proactive risk management.

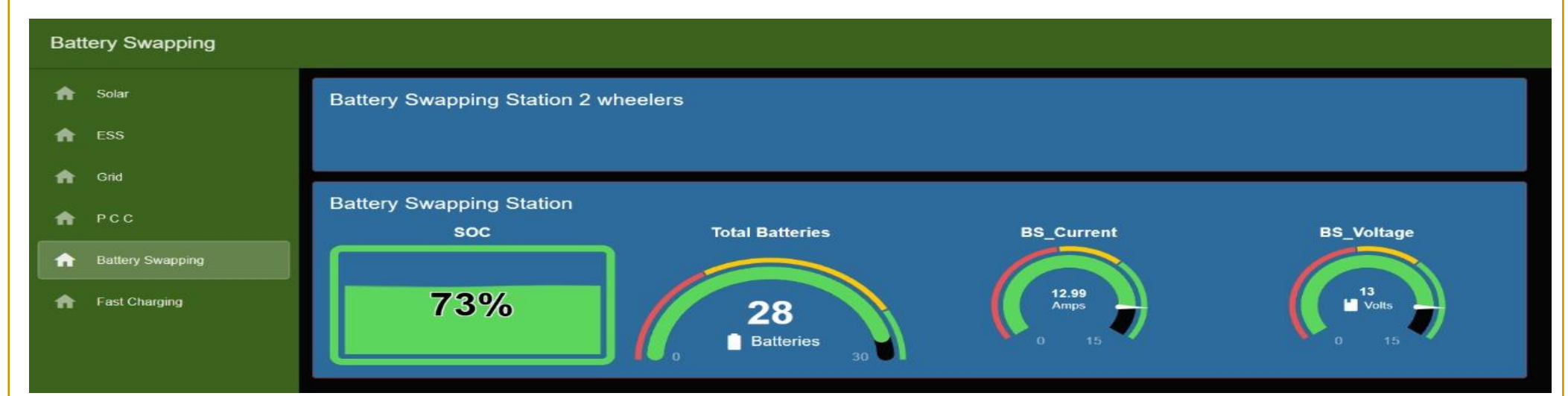
## Simulation work



## Real Time Monitoring



## Data Visualization



## Estimated Energy and Cost calculation

| Power Generation                                        |             |                          |                        |                          |                  |                                    |
|---------------------------------------------------------|-------------|--------------------------|------------------------|--------------------------|------------------|------------------------------------|
| Power Source                                            | Voltage [V] | Current [I]              | % of Generation/day    | Required Power           | Power [P] KW/hr  | Power/day                          |
| Solar                                                   | 430         | 16.7                     | 60                     | 72                       | 43               | 1032                               |
| ESS                                                     | 480         | 16.7                     | 30                     | 80                       | 24               | 576                                |
| Grid                                                    | 435         | 20.0                     | 10                     | 87                       | 9                | 209                                |
| Total Power Generation per day in KW/24 hr              |             |                          |                        |                          |                  | 1817                               |
| PCC - Point of common coupling                          |             |                          |                        | Battery Swapping Station |                  |                                    |
| PCC will logically controlled on availability of source |             |                          |                        |                          |                  |                                    |
| Voltage - [V]                                           | 430         | Output Power/day in KW   |                        |                          |                  |                                    |
| Current - [I]                                           | 17          | 75 1800 Gen/day          |                        |                          |                  |                                    |
| Power - [W] - KW/h                                      | 75          | 1.17 1772 use / day      |                        |                          |                  |                                    |
| Power Availability                                      |             | 1.17                     | 1772 use / day         |                          |                  |                                    |
| Fast Charging Station Load                              |             |                          |                        | ESS - Charging           |                  |                                    |
| Cars                                                    | Voltage [V] | Power [kW]               | Station Capacity [kWh] | SOC Available            | Required Current | Charging Time in %Cost charge Cost |
| 1- Tata Nano                                            | 400         | 36.2                     | 30.2                   | 20                       | 75.5             | 2.47 10 746.83                     |
| 2- MG ZS EV                                             | 400         | 50                       | 44.5                   | 30                       | 125              | 1.49 10 748.75                     |
| 3-Tata Tigor EV                                         | 400         | 15.8                     | 26                     | 50                       | 42               | 4.40 10 799.58                     |
| 4- Mahindra eVerito                                     | 400         | 70                       | 60                     | 10                       | 175              | 1.07 10 747.82                     |
| Total Power                                             | 167         | Total Time               |                        | 9                        | Total Revenue    | 2977                               |
| Total Power                                             | 1670        | KW power on charge / day |                        | Total Time               | / 96 hr          | Total Revenue 26794                |
| No. of batteries                                        |             |                          |                        | Voltage                  |                  |                                    |
| Amper hour - capacity                                   |             |                          |                        | 10                       |                  |                                    |
| Power / charge - Watt                                   |             |                          |                        | 100                      |                  |                                    |
| Power / one cycle charge - KW                           |             |                          |                        | 48000                    |                  |                                    |
| Power / one cycle charge - KW                           |             |                          |                        | 48                       |                  |                                    |
| Total Energy claculation                                |             |                          |                        |                          |                  |                                    |
| Power sources                                           | KW / day    |                          | KW / Month             |                          | KW / 5 years     |                                    |
| Required full Power from Grid                           | 1,812       | 54,510                   | 663,205                | 3,316,035                |                  |                                    |
| Solar                                                   | 1,032       | 30,960                   | 376,680                | 1,883,400                |                  |                                    |
| ESS                                                     | 576         | 17,280                   | 210,240                | 1,051,200                |                  |                                    |
| Grid                                                    | 209         | 6,270                    | 76,830                 | 384,135                  |                  |                                    |
| Total Power saving / Day                                | 1,608       | 48,240                   | 586,920                | 2,934,600                |                  |                                    |
| Total Generation Cost / Day \$/kw / unit                | 21,804      | 654,120                  | 7,958,460              | 39,792,300               |                  |                                    |
| Total Generation Cost Saving / Day, kw / unit           | 19,296      | 578,880                  | 7,043,040              | 35,215,200               |                  |                                    |
| Spending Amount                                         | 2,508       | 75,240                   | 915,420                | 4,577,100                |                  |                                    |
| Revenue                                                 |             |                          |                        |                          |                  |                                    |
| Source                                                  | Per day     | Per Month                | Per Year               | Per 5 Years              |                  |                                    |
| Fast charging                                           | 26,794      | 803,813                  | 9,779,719              | 48,898,594               |                  |                                    |
| Battery Swapping                                        | 2,700       | 81,000                   | 985,500                | 4,927,500                |                  |                                    |
| Total Revenue                                           | 29,494      | 884,813                  | 10,765,219             | 53,826,094               |                  |                                    |
| Spending Amount                                         | 2,508       | 75,240                   | 915,420                | 4,577,100                |                  |                                    |
| Total Profit                                            | 26,986      | 809,573                  | 9,849,799              | 49,248,994               |                  |                                    |

## Conclusion

The proposed IoT-based architecture addresses critical challenges in EV charging infrastructure by enhancing efficiency, integrating renewable energy, and providing real-time monitoring.

This innovation paves the way for sustainable transportation and significant cost savings.

## Contact

Jagadeeswara Chandra bose.k  
SRM University AP  
Email: [Jagadeeswara\\_kamarajugadda@srmmap.edu.in](mailto:Jagadeeswara_kamarajugadda@srmmap.edu.in)  
Phone: +91 9705233003

## References

- Cheng-Ta Chiang; Liang-Yu Chien "A Pitaya Dye-Sensitized Solar Cell Monitor for Environmental Sunlight Intensity Detection" IEEE Sensors Journal, 19-11, 03 February 2019.
- Saswat Kumar Ram; Sauvagya Ranjan Sahoo; Banee Bandana Das; Kamalakanta Mahapatra; Saraju P. Mohanty "Eternal-Thing: A Secure Aging-Aware Solar-Energy Harvester Thing for Sustainable IoT" IEEE Transactions on Sustainable Computing, 6-2, 15 April 2020.
- Tai-Yeon Ku; Wan-Ki Park; Hoon Choi "ESS Simulation based on PV Generation Prediction" 2018 International Conference on Information and Communication Technology Convergence (ICTC), 18 November 2018.
- Abdelrahman Ahmed; Sobhy Abdelkader; Ehab Ayman "IoT-based optimization of microgrid operation" 2023 24th International Middle East Power System Conference (MEPCON), 18 March 2024.
- George Y. Odongo; Richard Musabe; Damien Hanyurwimfura; Abubakar Diwani Bakari "An Efficient LoRa-Enabled Smart Fault Detection and Monitoring Platform for the Power Distribution System Using Self-Powered IoT Devices" IEEE Access (Volume: 10), 07 July 2022.
- Isa Hafidz; Ardiono Priyadi; Margo Pujiatara; Dimas Okky Anggriawan; Mauridhi Hery Purnomo "Development of IoT-Based Portable Power Quality Monitoring on Microgrids by Enhancing Protection Features" IEEE Access (Volume: 11), 15 May 2023.
- Dan Zheng; Fushuan Wen; Jiansheng Huang "Optimal planning of battery swap stations" International Conference on Sustainable Power Generation and Supply (SUPERGEN 2012), 04 April 2013.
- Bansilal Bairwa; Mrinal Sarvagya; Sanjeeva Kumar; Arpita Banik "Battery Swapping Mobile Service Station for Electric Vehicles Application" 2021 Second International Conference on Smart Technologies in Computing, Electrical and Electronics (ICSTCEE), 15 February 2022.
- Xian Zhang; Guibin Wang "Optimal dispatch of electric vehicle batteries between battery swapping stations and charging stations" 2016 IEEE Power and Energy Society General Meeting (PESGM), 14 November 2016.
- Sayantika Sikdar; Madhavi Damle "IoT Solutions for Electric Vehicles

